

Multimedia Technology

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Multimedia Technology

LEARNING UNIT 4

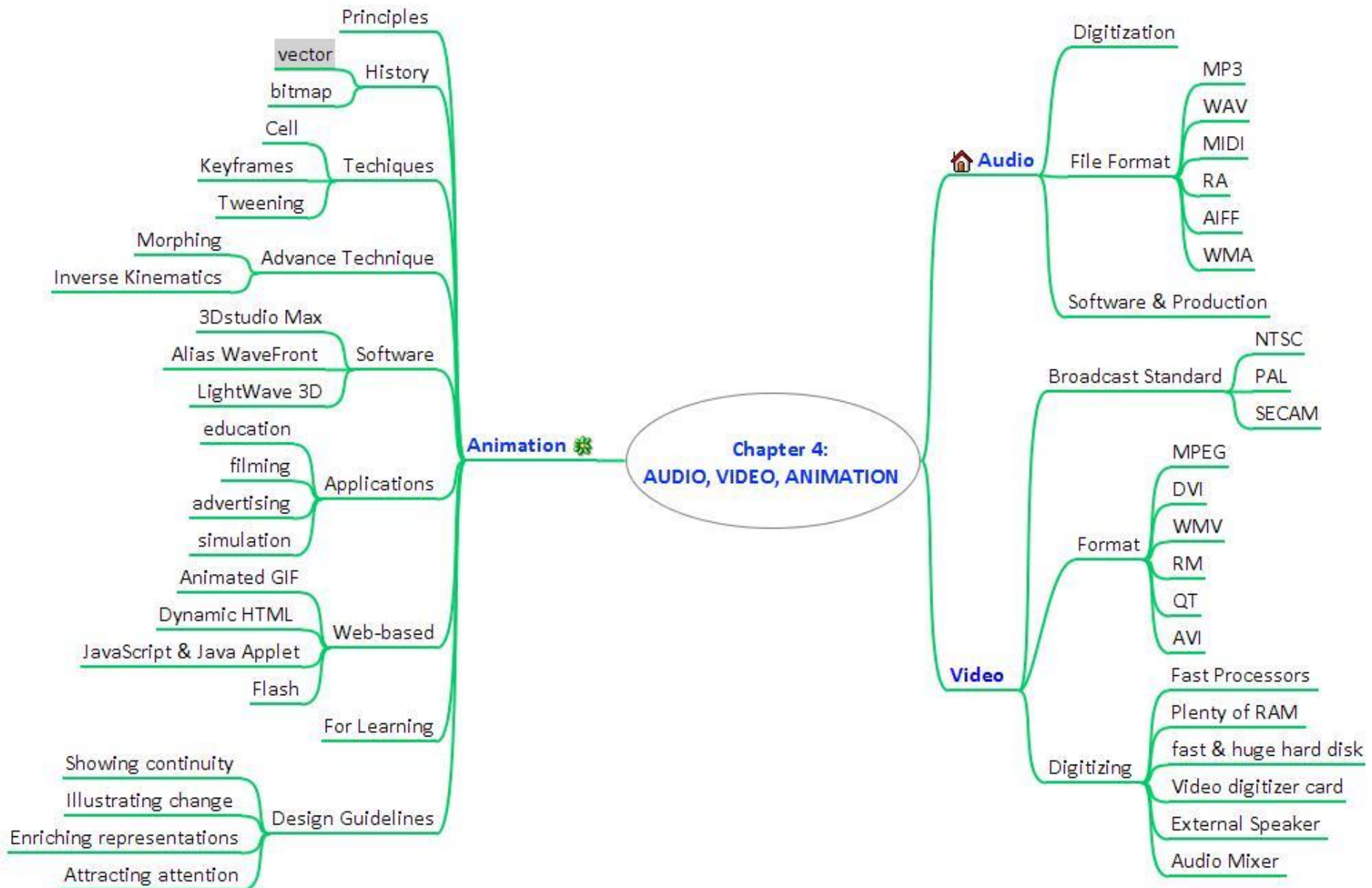
MULTIMEDIA : AUDIO, VIDEO AND ANIMATION

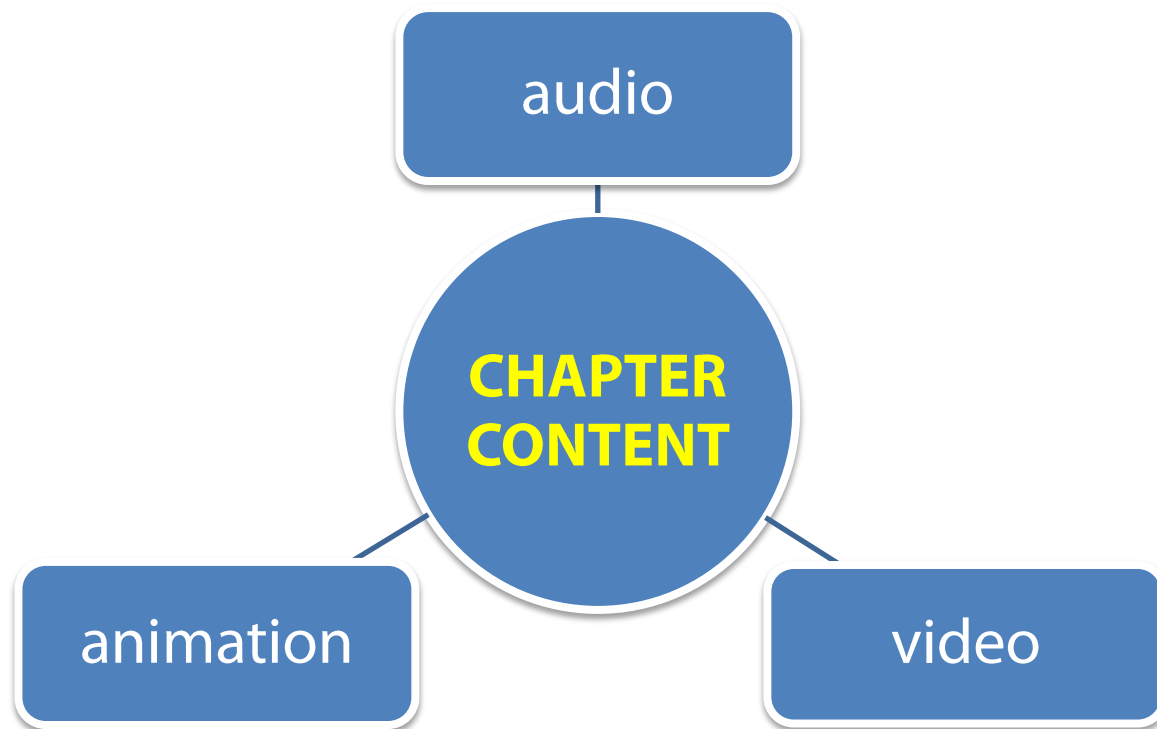


LEARNING OUTCOMES

- Discuss about audio digitization, audio file format and audio software.
- Explain about digital video standards, formats and technology.
- Examine basic principles behind animation and techniques.
- Discuss how animation is applied.

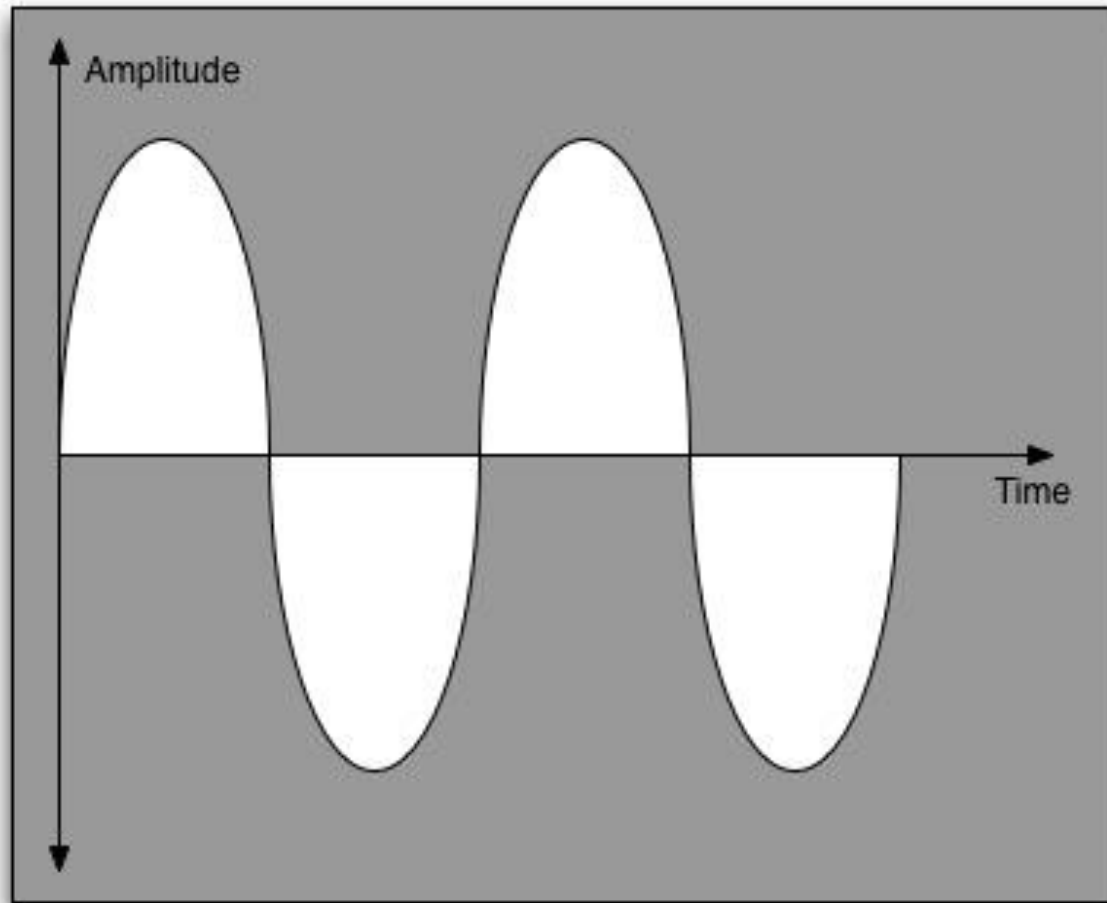


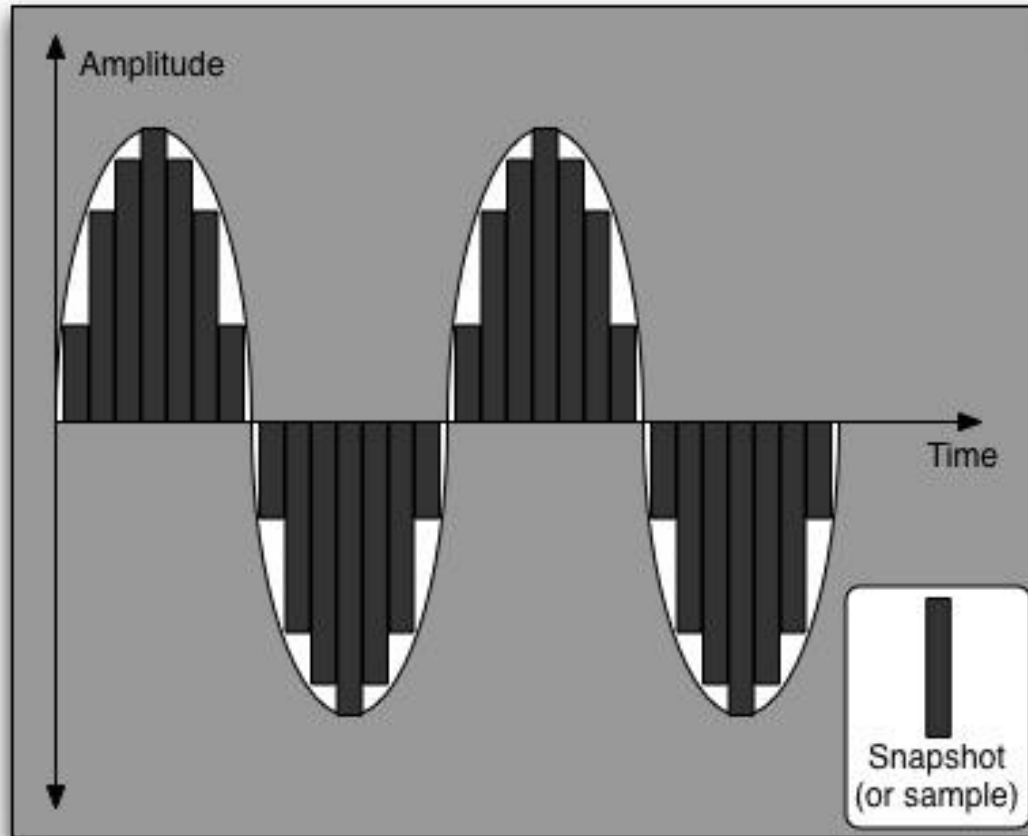


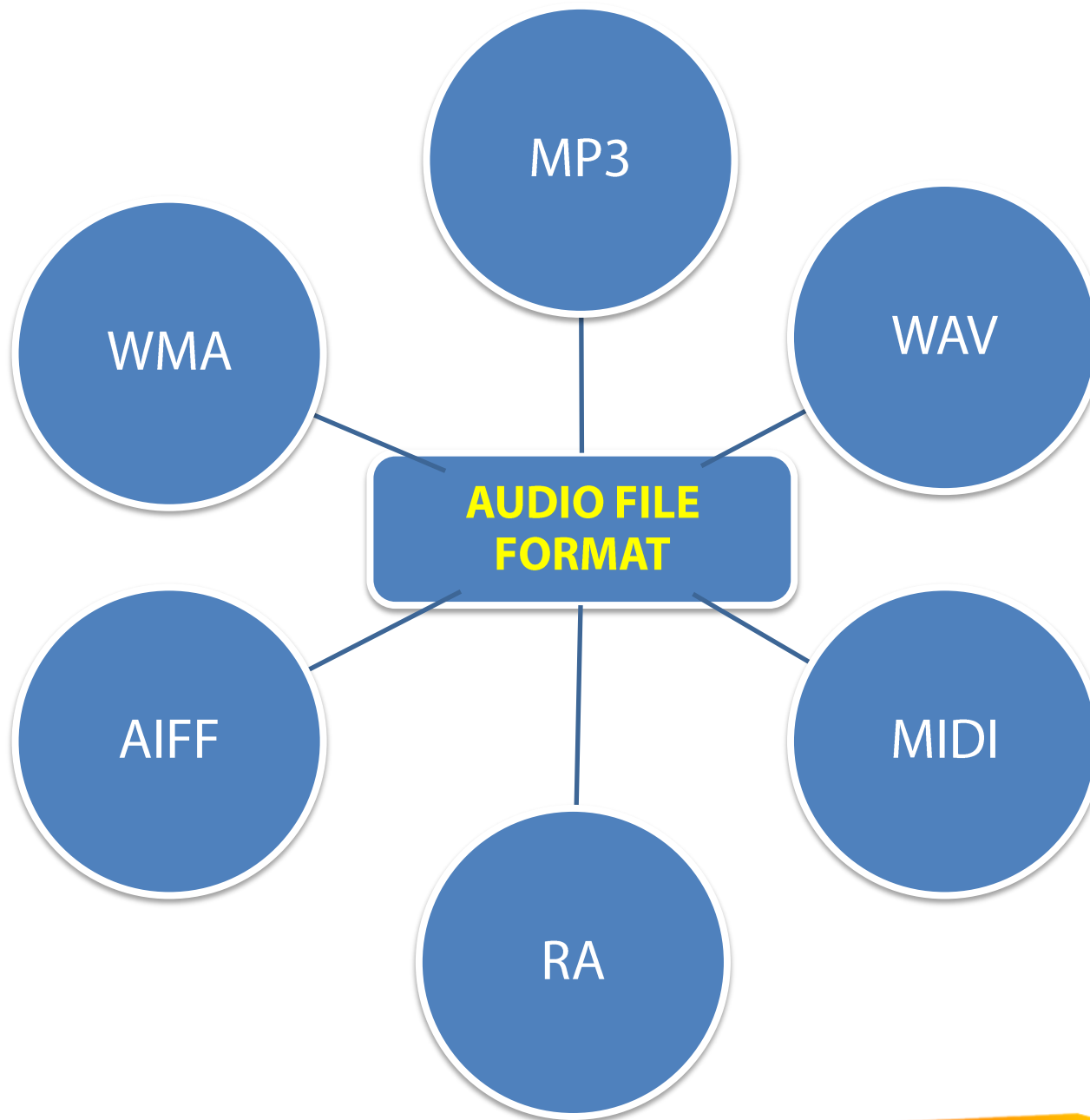


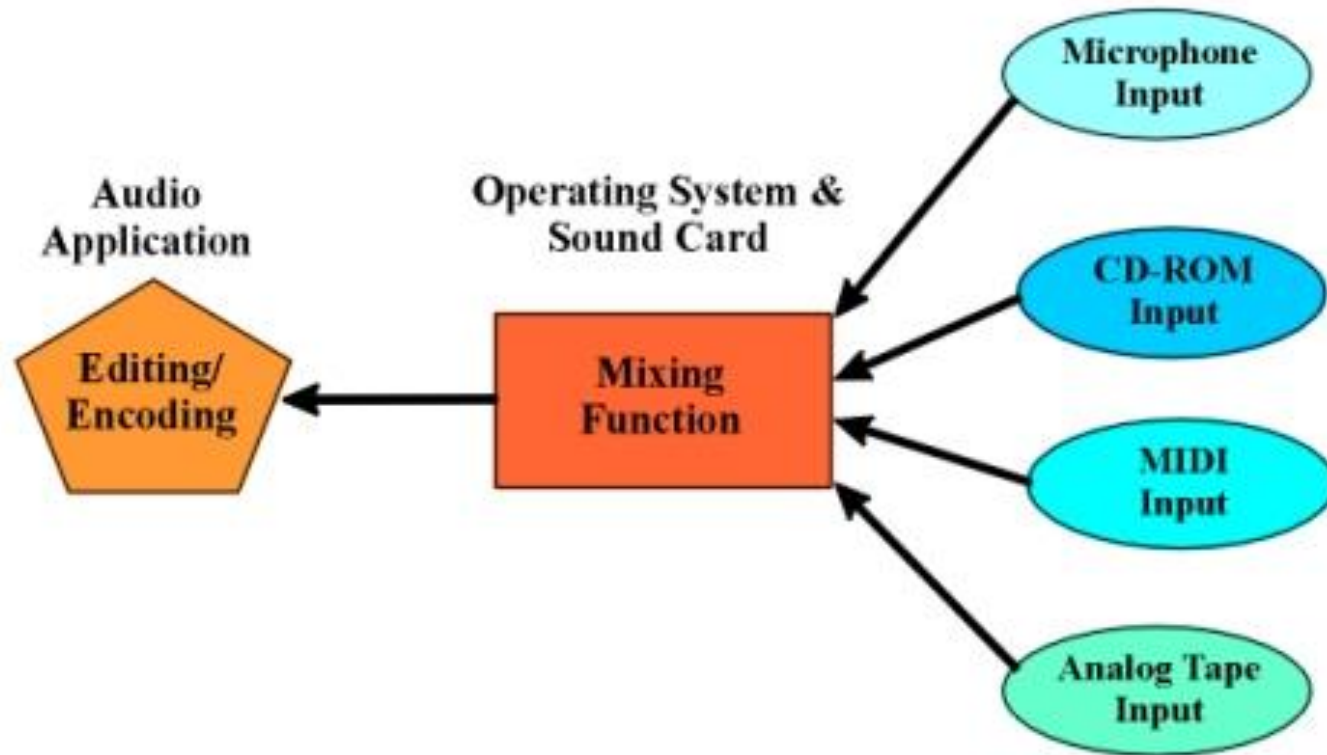
AUDIO



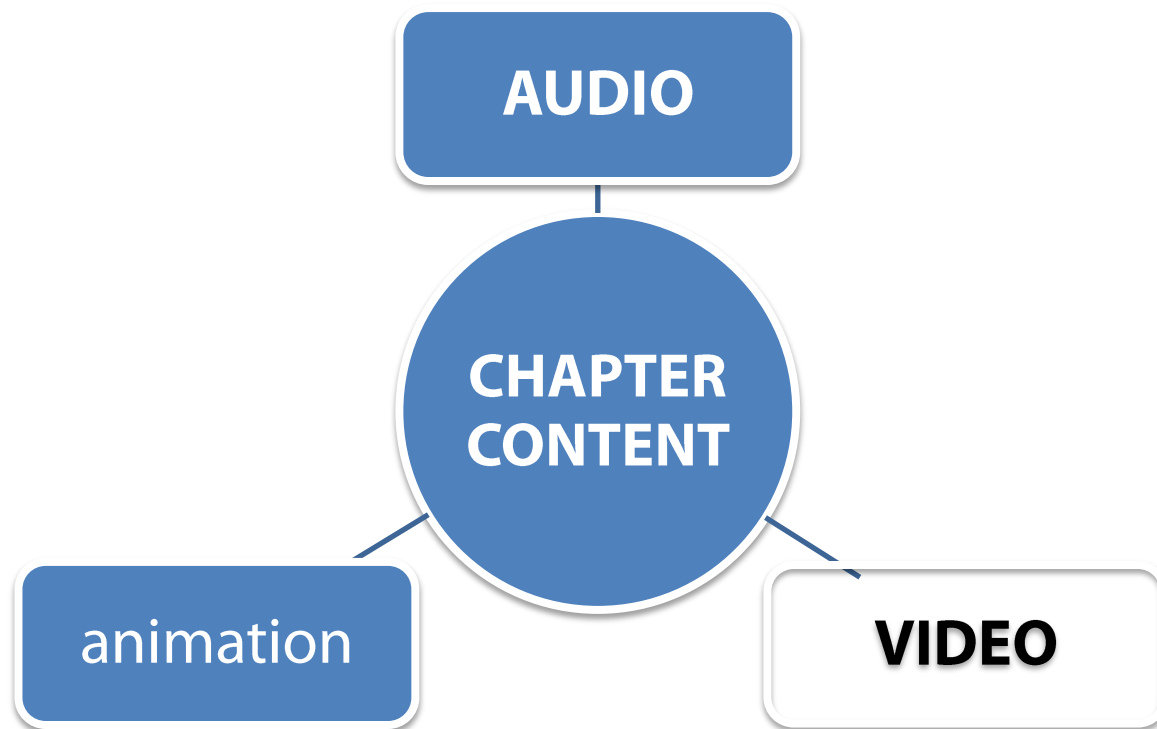








Quality	Sampling Rate (KHz)	Bits per Sample	Data Rate uncompressed	Freq. Band
Telephone	8	8 (Mono)	8 KBps	200-3,400 Hz
AM Radio	11.025	8 (Mono)	11.0 KBps	
FM Radio	22.050	16 (Stereo)	88.2 KBps	
CD	44.1	16 (Stereo)	176.4 KBps	20-20,000 Hz

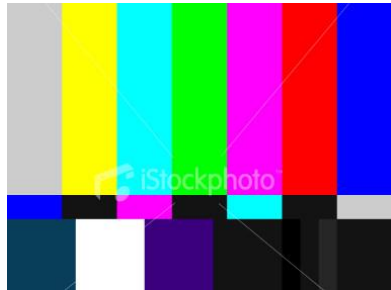


VIDEO

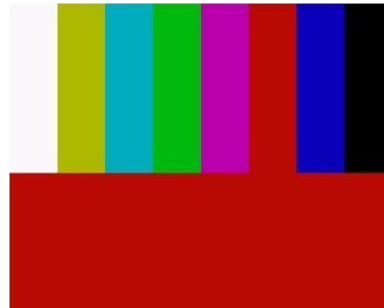


BROADCAST VIDEO STANDARDS

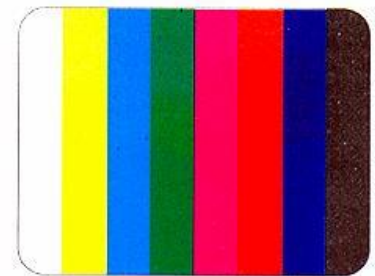
NTSC

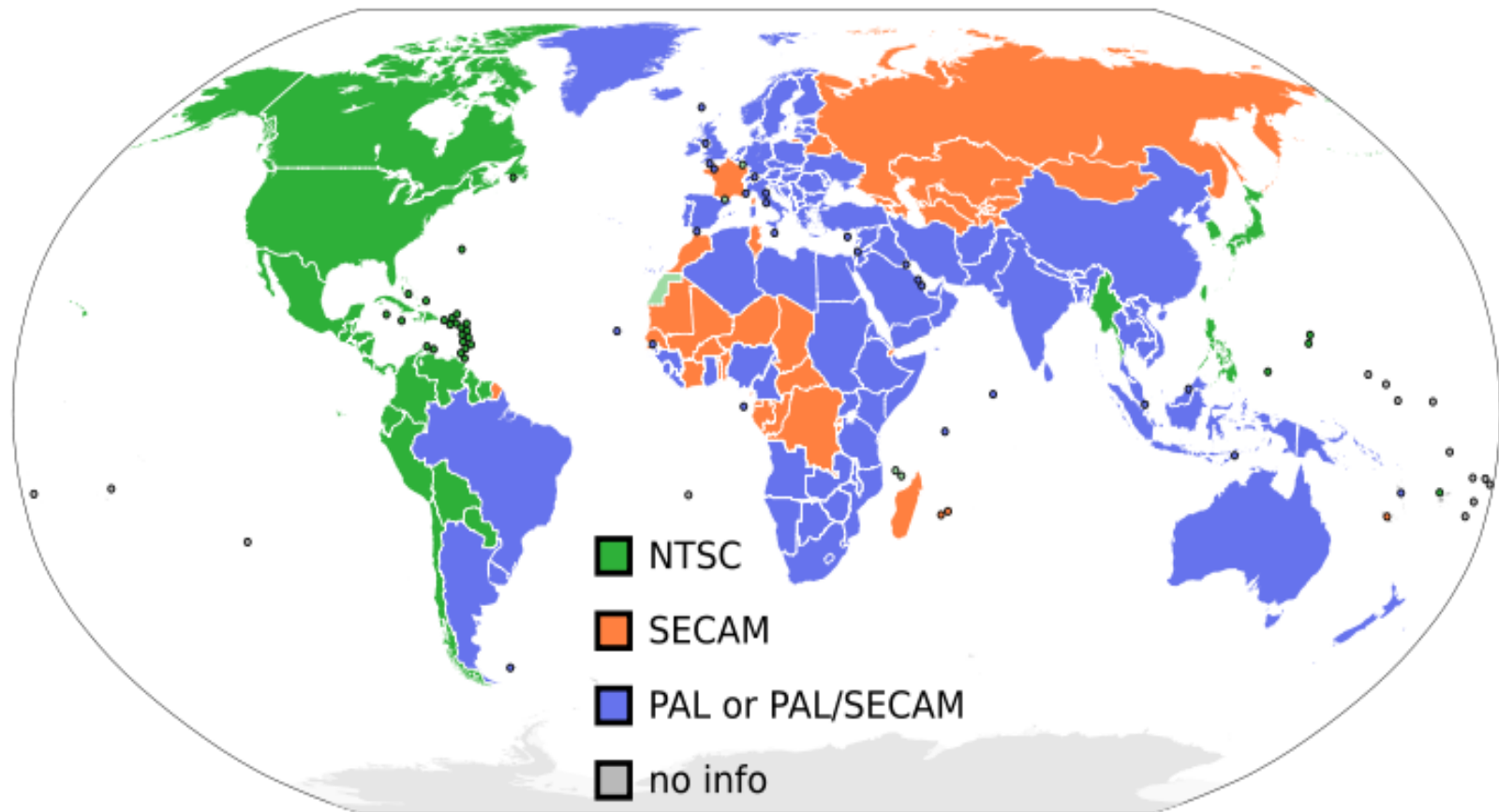


PAL



SECAM





NTSC

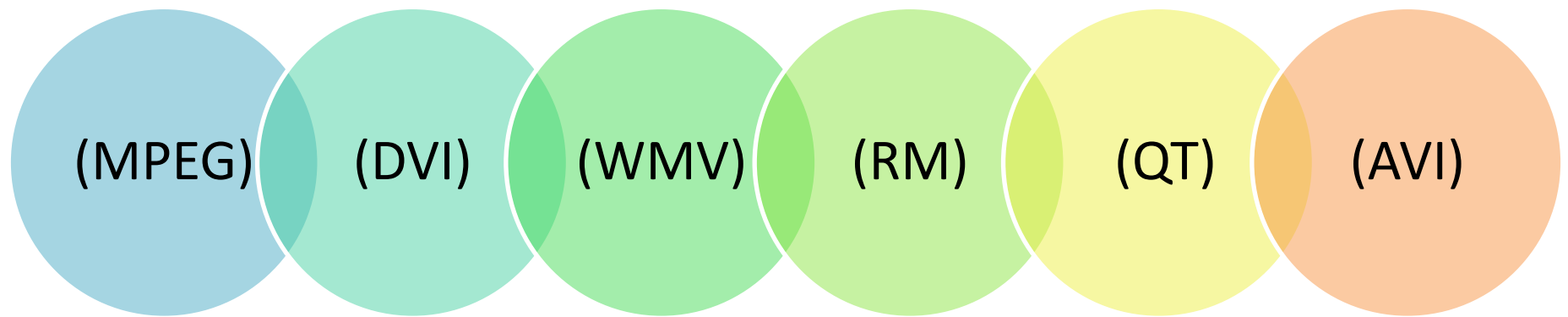
- used mainly in
 - North America
 - Japan
 - South America
- 30 fps (frames per second)
- Vertical - 525 scan lines
- Horizontal - 400 lines

PAL

- Phase Alternate Line
- 25 frames per second (fps)
- Vertical resolution 625 scan lines
- Used mainly in
 - Europe
 - China
 - Malaysia
 - Australia & New Zealand
 - Middle East
 - parts of Africa

SECAM

- SECAM - of Sequential Couleur Avec Memoire (sequential color with memory).
- 25 frames per second (fps)
- 625 lines per frame
- France and the former Soviet Republic



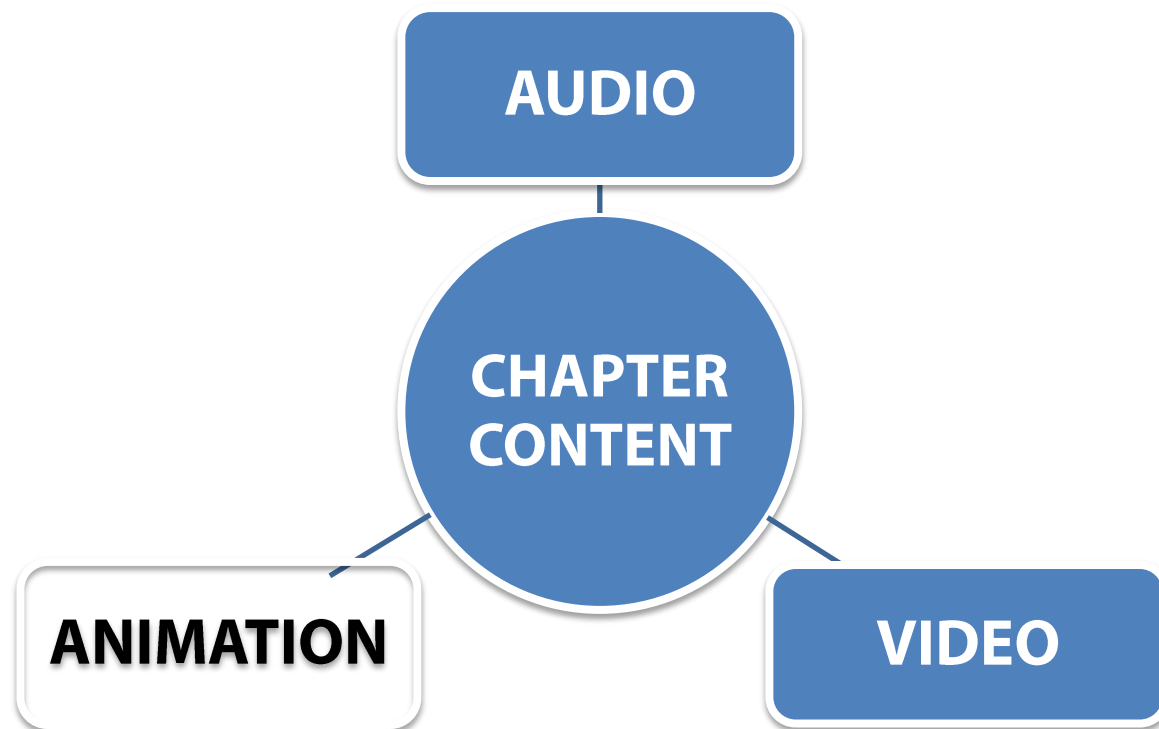
Fast Processors and
plenty of (RAM)

Fast and huge hard
disk space

**DIGITIZING VIDEO
- REQUIREMENTS**

External Speaker and
Audio Mixer

Video digitizer card
and video editing



ANIMATION



Contemporary and Forward Looking





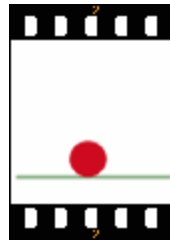
Contemporary and Forward Looking

Persistence of Vision

- pictures do not really move
- composed of a series of static images
- affect the eyes at the rate of 12 to 24 images per second.
- the illusion of movement is caused by a physiological affect known as 'persistence of vision'.

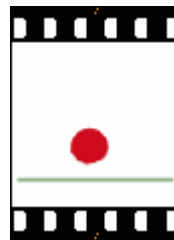


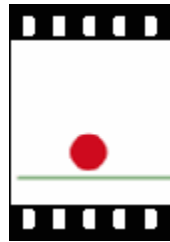


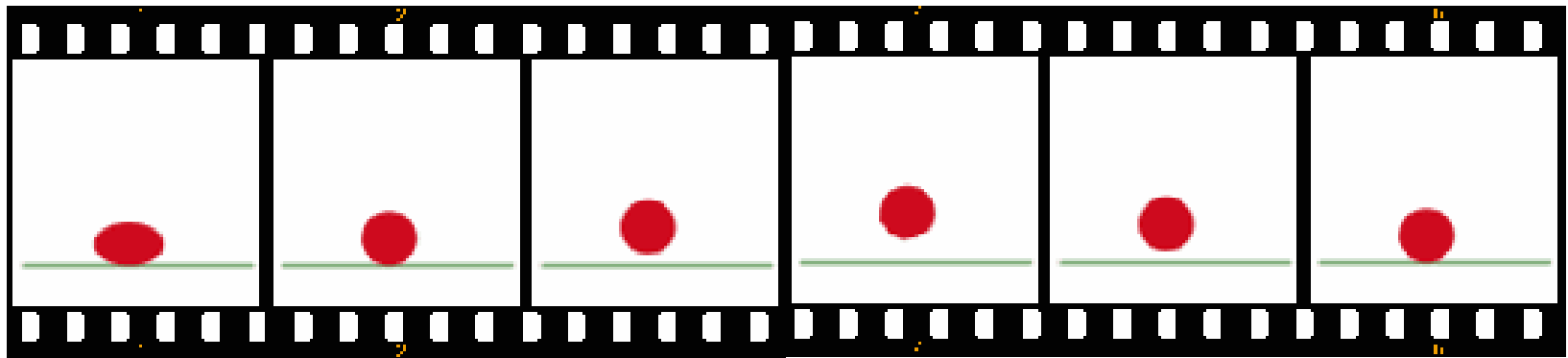












Animation Design Guidelines

- Showing continuity in transitions.
- Illustrating change over time.
- Enriching graphical representations.
- Attracting attention.



Summary

- MP3, WAV, MIDI, RA, AIFF, WMA are digital audio formats.
- NTSC, PAL and SECAM are broadcast video standards.
- MPEG, DVI, WMV, RM, QT, and AVI are digital video formats.
- Animation is an act of making something to come alive.
- Animation technology is applied widely in various application.
- Animated GIF, Dynamic HTML, JavaScript, Java Applet and Flash are examples of web-based animation technologies.
- In computer animation it is important to make sure that the animated character does follow the inverse kinematics law.

